

REGENSBURG OBERHUB, Germany

WMO: 107760

Lat: 49.0424N

Lon: 12.1020E

Elev: 371

StdP: 96.95

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.0	-11.1	-17.1	0.9	-12.7	-14.3	1.1	-10.2	10.9	5.8	9.5	4.6	2.1	0	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	29.7	18.7	27.7	17.9	26.1	17.3	20.0	26.9	19.1	25.2	18.3	23.8	2.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.5	21.7	17.1	12.8	20.9	16.3	12.2	20.2	58.9	27.2	55.9	25.2	53.2	23.8	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
8.0	6.5	5.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-15.8	33.3	4.5	2.3	-19.1	35.0	-21.7	36.3	-24.2	37.6	-27.5	39.2		
			-16.1	22.2	4.4	0.9	-19.2	22.8	-21.8	23.3	-24.3	23.8	-27.5	24.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	-1.3	-1.1	4.3	8.3	13.3	16.3	18.7	18.5	14.4	9.1	2.9	0.2
	DBStd	8.16	5.25	4.84	3.90	3.52	3.56	3.49	3.10	3.28	3.03	3.55	3.38	3.92
	HDD10.0	1505	352	310	180	74	13	1	0	0	2	60	212	302
	HDD18.3	3634	610	543	435	302	158	81	34	39	123	287	462	561
	CDD10.0	1029	0	0	3	22	114	189	269	265	134	32	1	0
	CDD18.3	117	0	0	0	0	1	19	45	46	6	0	0	0
	CDH23.3	1167	0	0	0	2	37	196	475	411	48	0	0	0
	CDH26.7	288	0	0	0	0	1	36	132	117	3	0	0	0
Wind	WSAvg	2.2	2.7	2.5	2.6	2.6	2.1	2.0	1.9	1.8	1.9	1.9	2.3	2.5
Precipitation	PrecAvg	636	45	36	40	39	61	79	72	73	47	48	51	49
	PrecMax	845	97	145	136	96	115	161	155	145	104	149	114	112
	PrecMin	449	19	8	3	11	16	16	23	22	11	4	8	5
	PrecStd	100	18	26	26	23	26	32	35	28	22	37	24	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	12.0	18.9	22.6	26.2	30.0	32.6	32.2	26.8	21.3	12.9	9.9
		MCWB	7.0	8.2	11.2	12.6	15.8	18.8	19.8	19.1	17.5	14.7	10.2	8.2
	2%	DB	7.9	8.4	15.0	19.9	24.3	27.5	29.7	29.8	24.8	18.2	11.0	8.3
		MCWB	5.8	5.7	8.8	11.6	14.6	17.7	19.0	18.6	17.2	13.3	9.6	6.9
	5%	DB	6.4	6.4	12.1	17.6	22.8	25.6	27.9	27.5	22.6	16.1	8.9	7.0
		MCWB	5.1	4.5	7.5	10.4	13.9	17.1	18.1	17.8	16.3	12.5	7.2	6.0
	10%	DB	4.8	4.7	9.9	15.4	21.0	23.6	26.1	25.5	20.4	14.2	7.5	5.5
		MCWB	3.8	3.2	6.6	9.6	13.1	16.2	17.5	17.2	14.8	11.6	6.2	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	8.4	11.9	13.7	17.6	20.7	21.6	20.9	19.2	16.2	11.0	8.7
		MCDB	9.6	11.6	18.3	20.5	22.7	26.8	28.5	29.0	24.4	19.8	12.0	9.5
	2%	WB	6.4	6.3	9.5	12.3	15.9	19.4	20.2	19.6	18.1	14.3	9.5	7.2
		MCDB	7.6	8.0	13.2	18.4	21.0	24.9	26.9	26.9	23.0	17.0	10.9	8.0
	5%	WB	5.1	4.7	8.2	11.1	14.9	18.2	19.3	18.7	17.1	12.9	7.5	6.0
		MCDB	6.2	5.9	11.0	16.3	20.3	23.5	25.4	24.8	20.8	15.0	8.5	7.0
	10%	WB	3.7	3.3	7.1	10.1	14.0	17.0	18.4	18.0	15.9	11.9	6.3	4.6
		MCDB	4.8	4.5	9.8	14.3	20.0	22.1	24.0	23.2	19.2	13.9	7.2	5.4
Mean Daily Temperature Range	5% DB	MDBR	4.6	6.3	8.2	10.3	11.2	10.5	11.6	11.2	10.0	8.0	4.9	4.1
		MCDBR	6.0	7.5	12.9	15.0	14.7	13.9	15.3	15.0	13.1	10.7	6.9	5.3
		MCWBR	4.6	5.1	7.0	7.2	6.1	5.6	5.8	5.4	6.2	6.2	5.0	4.5
	5% WB	MCDBR	5.7	7.2	11.1	13.3	12.3	12.4	12.7	12.3	11.2	9.6	6.1	5.2
		MCWBR	4.6	5.3	6.5	6.7	5.8	5.5	5.6	5.1	5.8	6.2	4.9	4.6
Clear-Sky Solar Irradiance	taub	0.303	0.325	0.362	0.396	0.403	0.405	0.413	0.407	0.380	0.354	0.328	0.300	
	taud	2.417	2.367	2.320	2.260	2.272	2.296	2.269	2.307	2.358	2.421	2.454	2.439	
	Ebn at Noon	758	821	844	850	858	856	842	830	816	776	720	711	
	Edh at Noon	65	85	106	124	128	126	128	118	101	81	63	56	
All-Sky Solar Radiation	RadAvg	0.92	1.80	2.88	4.27	5.00	5.53	5.28	4.63	3.32	2.02	1.00	0.71	
	RadStd	0.15	0.26	0.37	0.55	0.58	0.47	0.50	0.46	0.43	0.24	0.12	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (60 neighbors)	+0.39	N/A	N/A	N/A	N/A	+0.35	N/A	N/A	+55	+20

Nomenclature: See separate page